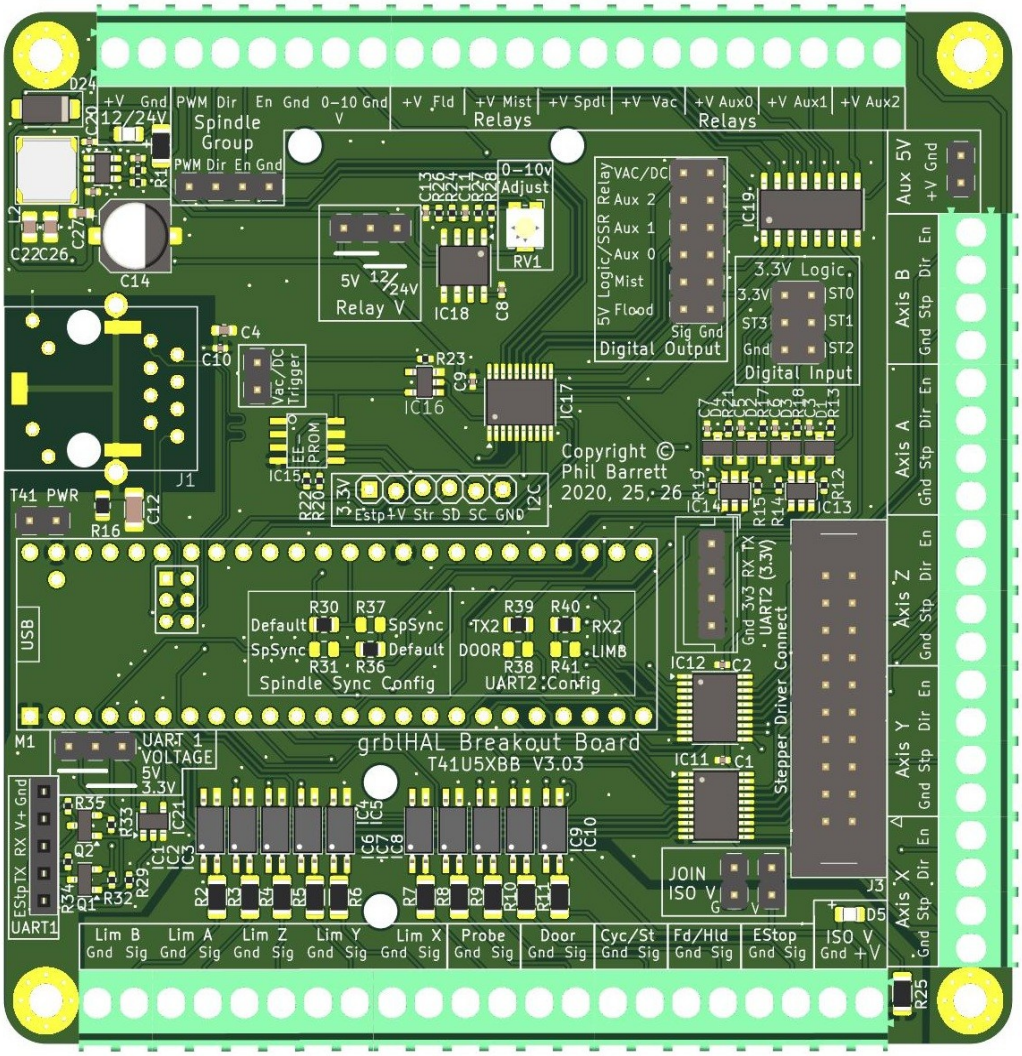
grblHAL T41U5XBB Teensy 4.1 Motion Controller
User Manual

Version 3.0x



Date 10/20/2026

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Introduction

The T41U5XBB when paired with grblHAL creates a powerful 5 Axis motion controller system.

New in V3.0x

New features in Version 3

- Supports 12V to 24V input
- Onboard 5V regulator for both on and off board use
- Reversed power polarity protection
- Fully isolated Limit, Probe and Control inputs
- UART1 has 3.3V/5V voltage translation and dedicated ESTOP pin
- Second UART, board level configuration
- Spindle Sync board level configuration
- Alternate 2.54mm pin header footprints for the screw terminals
- Ground plane exposed to mounting screws

Feature Overview

- 5 Axis control outputs
  - 5V compatible.
  - Step, Direction and Enable.
  - Screw Terminal and Pin Header.
  - Independent Enables for all axes.
- Limit Inputs for all axes
  - Opto-isolated.
  - Supports switch and powered proximity sensors.
  - 12/24V for better EMI immunity.
  - Separate 12/24V power supply for full isolation.
- Standard Grbl control inputs
  - Opto-isolated.
  - Cycle Start, Feed Hold, Safety Door, Reset.
  - 12/24V for better EMI immunity.
- Relay Support
  - Directly drive up to 7 Relay coils: Spindle enable, Flood, Mist, Dust Collection, 3 auxiliary outputs.
  - Relay Coil drive voltage selectable – 5V or 12/24V.
  - Drive up to 7 SSR or Logic Level relay modules: Spindle enable, Flood, Mist, Dust Collection, 3 auxiliary outputs.
- Dust Extraction
  - Direct relay support. Activated by spindle enable.
  - Logic level trigger.
  - Separate control of Dust Collector relay via pin header.
- Relay coil voltage switchable between 5V and 12 or 24V.
- Spindle/Laser support
  - Standard GRBL spindle control outputs (5V): Enable, Direction.
  - 0-10V spindle control with output level adjust.
  - PWM (5V).
- Screw Terminals for reliable connections available for most I/O.
  - Alternate 2.54mm pin header footprints for each screw terminal.
- Digital Inputs
  - 4 Inputs.
  - EMI filtered, 15.9KHz low pass, Schmitt Trigger.
  - Support for grblHAL Spindle Sync.

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- Flexible power architecture
  - 12V or 24V main board power input.
  - 12V or 24V isolated section power input.
  - Main and Isolated inputs can be joined and powered by a single PSU.
  - On board 5V generation.
  - 5V Logic section may be alternately power by USB.
  - Auxiliary 5V output up to 500 mA.
  - LED indicators for power input and isolated power input.
- Communications
  - USB interface (on user provided Teensy 4.1).
  - Ethernet interface.
  - 2 UARTs
    - UART1 Header with 3.3V/5V translation and ESTOP pin.
    - UART2 Header, 3.3V only.
  - I2C header with signal and ESTOP pins.
- Micro SD card socket (on user provided Teensy 4.1).
- EEPROM footprint for SOIC8 devices.

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Feature details

Stepper Driver outputs.

There are 5 sets of standard GRBL stepper driver outputs: X, Y, Z, A and B axes. Each has 3 pins: step, direction and enable. Each are 5V compatible and capable of driving opto-isolated stepper driver inputs. Each axis has a set of screw terminals as well as pin headers (0.1”/2.54 mm spacing).

Limit Pins.

There are 5 limit inputs: X, Y, Z, A and B. All are opto-isolated and available via screw terminals. These may be powered via an isolated 12/24V PSU. They may also be powered from the main power 12/24V input by joining the two power sections.

Control Input Pins.

Standard GRBL control input pins are available via screw terminals: Feed/Hold, Cycle/Start, Halt/Stop and Safety Door open. These are opto-isolated. These may be powered via an isolated 12/24V PSU. They may also be powered from the main power 12/24V input by joining the two power sections.

Probe Input

Probe input is opto-isolated and available via screw terminals. This may be powered via an isolated 12/24V PSU. They may also be powered from the main power 12/24V input by joining the two power sections.

Spindle Control.

Outputs for Spindle Enable, Spindle Direction and PWM signals available via screw terminal and pin header. These are 5V TTL logic compatible. There are relay outputs for the spindle (direct relay coil and 5V TTL). See Relay section for more details. In addition, standard 0-10V output for spindle control is provided via a screw terminal.

Relay outputs.

Standard GRBL Flood and Mist relay outputs are supported via screw terminals. Spindle relay output via a screw terminal is supported and controlled by Spindle Enable. In addition, a separate screw terminal relay output for Dust Collection is activated by the Spindle Enable output.

3 auxiliary relay outputs are supported via screw terminals and used via M62-M65 grblHAL pin control.

All relay coil outputs are capable of directly driving relay coils. Each relay driver is capable of up to 100 mA. All 7 are limited to 500 mA in total. Above 250 mA, forced air cooling is recommended.

All relay coil drivers have open collector outputs.

Each relay also has a 5V TTL output that can be used to drive solid state relays, separate relay boards or other low voltage activated devices.

Relay coil voltage is selectable between 5 and 12V/24V via the pin header labeled Relay Voltage in the upper left corner of the PCB or, alternatively, via a solder jumper on the bottom the PCB.

Schmitt Triggered digital inputs:

Four digital inputs are provided via a pin header. These have Shottky Diode protection and a low pass RC filter with an 15.9 KHz cutoff frequency. See the schematic for the specific circuit.

I2C Header

An I2C header is provided. 3.3V only, there is no translation to/from 5V and the inputs are not 5V tolerant. A strobe line from the Teensy 4.1 pin 33 is provided on the header. In addition, a line for an E-Stop signal is routed directly to the grbl Halt input. Currently, only a limited number of devices are supported. Check the grblHAL github site for more details.

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USB

USB is connected via the user provided Teensy 4.1.

Ethernet

If the optional kit is installed, Ethernet may be connected via the RJ45 jack.

UART

Two UARTs are supported. UART1 has selectable voltage (3.3V or 5V) and an additional ESTOP input pin.

Power Inputs

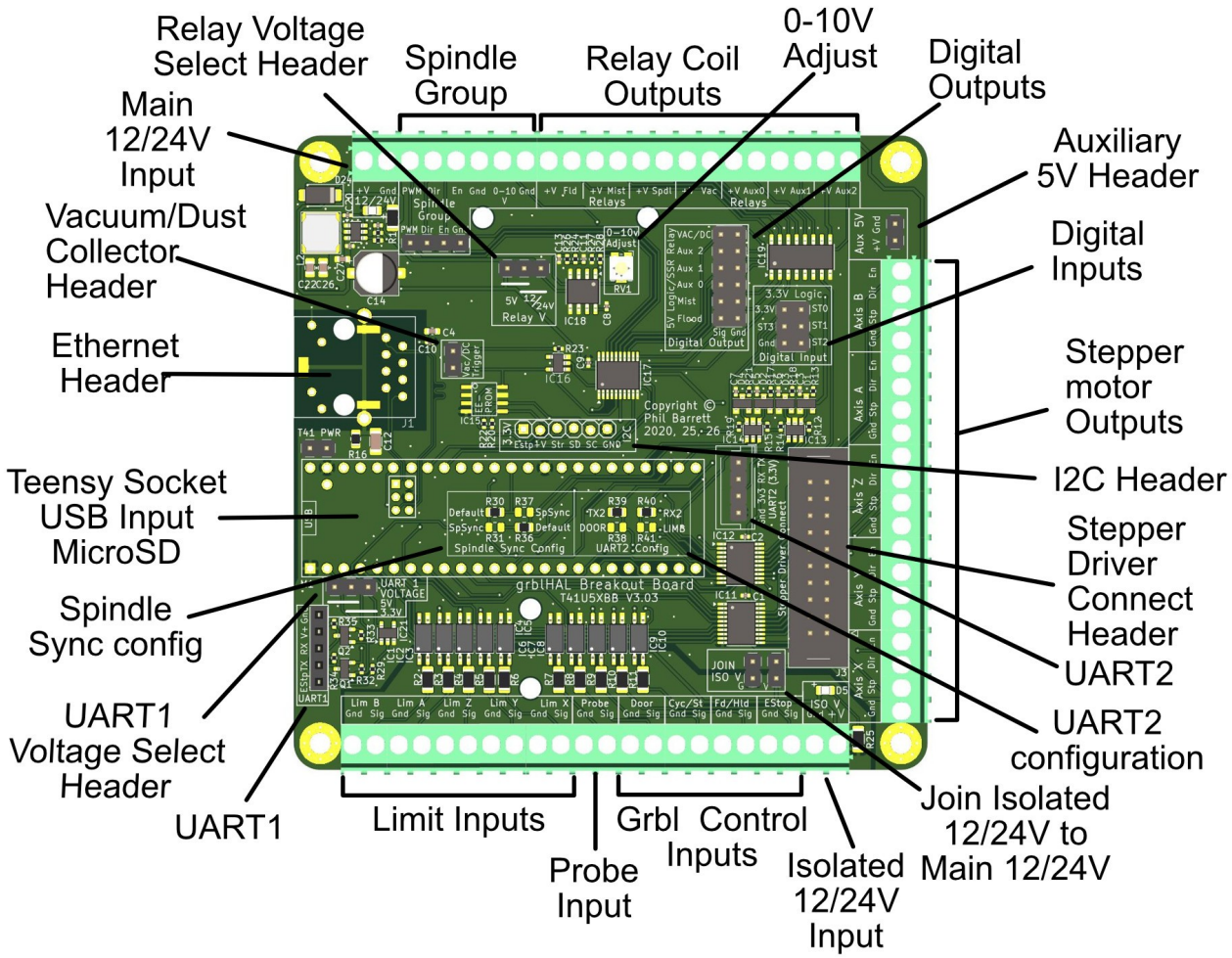
The T41U5XBB is powered via a 12V to 24V input. There are two sets of screw terminals, one for the main board power and one for the isolated section power. They may optionally be powered from the main power 12/24V input by joining the two power sections. In that case, the isolated section power connector may be used to power 12/24V devices.

Auxiliary 5V output

5V is available via the AUX 5V header for the system integrator. The onboard 5V regulator can deliver up 500 mA. The footprint supports either a 2.54mm pin header or a JST XH (or similar) header.

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External Connections



Main 12/24V input
: Supply for on-board 5V generation, relays and 0-10V output amplifier.

Isolated 12/24V input
: May be different from Main 12/24V input.

Auxiliary 5V Output
: Supply for external 5V usage.
: Can supply up to 500 mA to off board devices.

UART1
: Voltage translation (3.3V/5V) as selected by the header.

UART2
: 3.3V only.

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Stepper motor outputs

All signals are 5V logic.

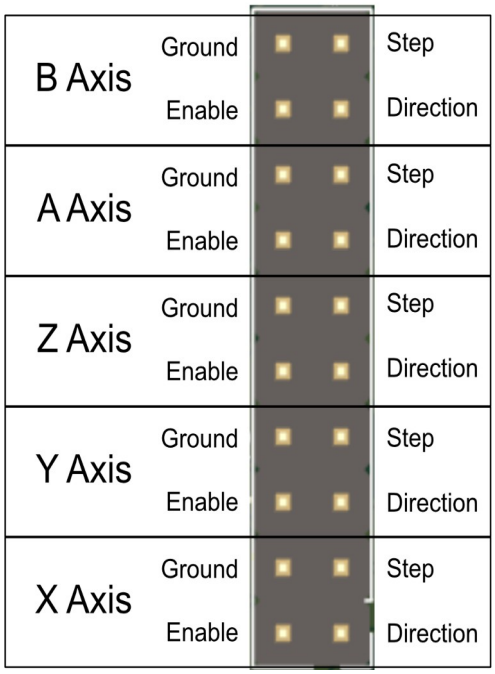
Pin	Signal
G	Ground
Stp	Step Pulse
Dir	Direction
En	Enable

Digital outputs

These are 5V, TTL compatible outputs suitable for SSR and powered relay modules.
Flood, Mist, Vac/Dust Collector, Aux0, Aux1, Aux2.
Note: Enable in the Spindle Group is capable of driver SSRs and powered relay modules.

Stepper Driver Connect Pin Header

All signals are 5V Logic.



Limit Inputs

One input for each axis. Opto-isolated, 12V or 24V depending on user supplied power. Open Collector/Drain compatible.

Grbl Control Inputs

Halt, Door, Cycle/Start and Feed/Hold. Opto-isolated, 12/24V depending on user supplied power. Open Collector/Drain compatible.

Probe Input

Opto-isolated, 12/24V depending on user supplied power. Open Collector/Drain compatible.

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Spindle Group, Output.

PWM, Direction, Enable, Ground. 5V logic..
0-10V signal.

Relay Coil Outputs

These are open collector/drain sinks. They run at the selected relay voltage.
Flood, Mist, Spindle, Vac/Dust Collector, Aux0, Aux1, Aux2

Vacuum/Dust Collector Header.

External Switch to activate the vacuum or dust collector relay.

Digital Input

4 input pins, 3.3V and Ground. Schmitt Triggered input, diode protected and low pass filtered. See schematic for more details.

Pin	Signal
ST0	Schmitt Triggered, Low Pass Filtered (15.9KHz)
ST1	Schmitt Triggered, Low Pass Filtered (15.9KHz)
ST2	Schmitt Triggered, Low Pass Filtered (15.9KHz)
ST3	Schmitt Triggered, Low Pass Filtered (15.9KHz)
3.3V	3.3VDC
Gnd	Ground

ST0-ST3 are 5V tolerant.

I2C Header, Input/Output.

These are 3.3V only and not 5V tolerant.

Pin	Signal
G	Ground
SC	I2C SCL
SD	I2C SDA
St	Strobe
+V	3.3VDC, source by Teensy 4.1
Hlt	Halt Signal. Connected to the Halt input Teensy 4.1 pin. 3.3V, no translation.

Ethernet Header

Requires MagJack. 10/100 MBPS.

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Assembling the T41U5XBB

<update required. Photo session with V3.03 board needed>
Not all components are required. See Optional Components section for more information.

Preparing the Teensy 4.1

Materials for socketing. See BOM section for ordering numbers.

- 2 24 pin 0.1” pin-header strips (snap off of 1x40 header strip).
- 2 24 pin 0.1” sockets.
- 1 riser 2x3 2 mm pin header. (For Ethernet)
- 1 2x3 2 mm socket. (For Ethernet)

See the blog post about building the T41U5XBB in pictures.

Before installing components, you need to decide how you will connect the Teensy 4.1 to the breakout board. There are two approaches:

1. Use sockets to mount the Teensy to the breakout board. This is the preferred approach. It is more flexible but runs the risk of shaking loose in a moderate to high vibration environment.
2. Directly solder the Teensy to the breakout board. This is the recommended approach for high vibration environments. It makes for a much more secure connection. This is also easier than the socket approach if using Ethernet.

Assembling the T41U5XBB

The T41U5XBB is provided with all surface mount components soldered in place. You are responsible for installing the through-hole connectors, either from the option assembly kit or sourced yourself.

Materials for basic board assembly

- Prepared Teensy 4.1 (see above).
- 5 1x2 2.54mm headers (snap off of 1x40 header strip).
- 2 1x3 2.54mm headers (snap off of 1x40 header strip).
- 2 1x4 2.54mm headers (snap off of 1x40 header strip).
- 1 1x5 2.54mm headers (snap off of 1x40 header strip).
- 1 2x3 2.54mm headers (snap off of 1x40 header strip).
- 1 2x6 2.54mm headers (snap off of 1x40 header strip).
- 32 3.5mm screw terminals.

Material for Ethernet assembly

- MagJack (see Bill of Materials section for exact part number, no substitutions).
- 1 2x3 2mm header.
- 1 2x3 2mm socket.

Materials and tools for general soldering

- Solder (Lead free or Tin-lead). 0.5mm to 1mm diameter.
- Soldering iron with a fine tip. Chisel tip 1mm to 2mm wide. The 1mm tip is better for the 2mm header.
- Solder wick (optional, for cleaning up solder bridges.

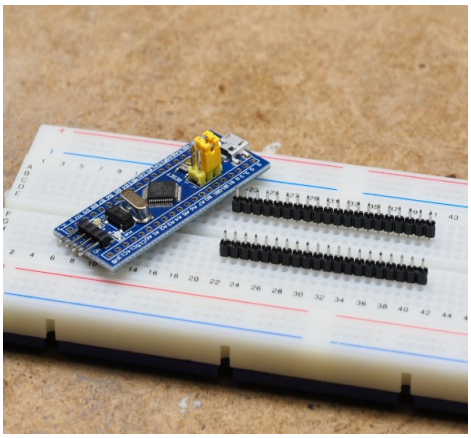
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Assemble the PCB in the following order:

1. Install 1x2, 1x3, 1x4 and 1x5 pin headers: Join ISO V (2 1x2),Vac/DC Trigger (1x2), Relay V selection (1x3), Spindle Group (1x4) UART 1 (1x5), UART 1 Voltage (1x3) and UART 2 (1x4). Carefully snap each one off the longer pin-header strip. Use tape (painter's tape works well) to hold them in place. Solder 1 pin and then verify the alignment. The header should be perpendicular to the board. If it is not, reheat the soldered pin and gently press it into place (watch the heat on the short headers!). Be sure to be gentle with heating headers as the plastic spacer softens quickly.

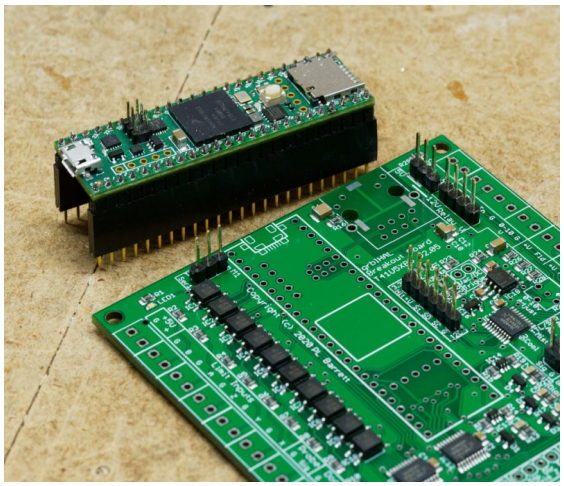
2. Install 2x3 and 2x6 pin headers. Follow the above procedure for soldering.

3. If you bought a Teensy 4.1 without headers, install pin headers. A solderless breadboard makes a great way to stabilize the pin headers while soldering the 0.1” header strips. (The photo below is just for illustration, that is not a Teensy.)



If using Ethernet, solder the 2x3 2 mm header in place on the bottom of the teensy 4.1. Use tape to hold it in place, solder 1 pin and check for alignment. If misaligned, reheat the soldered pin and press the header into correct alignment. Then, solder the other pins.

4. Install the Teensy 4.1. Place the sockets on the pin headers you just soldered onto the Teensy 4.1. Note, in the picture below, the 2x3mm pin header on the Teensy 4.1 is for testing purposes and not needed for the USB version. The Ethernet version uses a 2x3mm header that is attached to the underside of the Teensy.



Now, gently insert the Teensy 4.1 assembly into the indicated holes on the breakout board. Verify that all socket pins are correctly in place. Secure the entire assembly with tape on the component side of the board and then solder one pin on each of the different sockets and verify alignment. Once you are sure that the alignment is correct, solder the other pins, leaving the pins you initially soldered to last.

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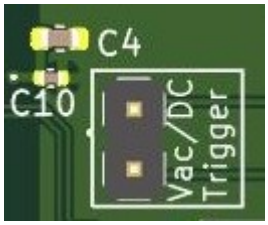
5. Mount and solder the 3.5mm screw terminals. Assemble a complete row before soldering. The terminals lock together but are sometimes hard to get completely aligned. If that is the case, place them upside down (with the pins pointing up) on a hard surface and press each terminal in place with a screwdriver or similar. When you are satisfied that the tops are flush, insert and solder an end pin. Then solder the pin at the other end, making sure the row is properly aligned. It is sometimes helpful to solder the pin of middle a middle terminal while moving the row into proper alignment. Then solder remaining terminals while ensuring that the row is straight

Optional Components.

Many components are optional.

- In general, we recommend that you install all screw terminals at the beginning. Adding them later can be difficult due to their interlocking nature.
- UART 1, UART 2 and I2C headers are optional. grblHAL will work correctly without either header installed. Specific options may need them.
- EEPROM is optional, Install if you need to use an external EEPROM. grblHAL will work correctly without it.
- 0-10V Spindle control is optional. Use if you are using a VFD controlled spindle. grblHAL will work correctly without it if you are manually controller your spindle. We recommend installing the screw terminals as they are difficult to add later.
- Relays are optional. The T41U5XBB is capable of driving relay coils directly. It is recommended that you install the screw terminals even if you don't initially plan on using them. Installing later can be difficult. See note on Vacuum/Dust Collector relay below. But, grblHAL will work correctly without any relays.
- Safety Door alarm. Install if you have an enclosure for your CNC machine with a Safety Door sensor. You will need to build and load a version of grblHAL with it enabled. You will need to change the board lever Door configuration. By default, the Web Builder does not configure Safety Door.
- Stepper Driver Connect. This is a 2x20 2.54mm header that supports each Axis in 5 groups of 4 pins. Install these only if you intend on using a them to connect to your stepper drivers. For reliability, screw terminals are recommended over pin headers. You can install a shrouded header for the SDC and use a IDC connector with ribbon cable.
- Ethernet MagJack. If you are not using Ethernet, there is no need to install this. If using Ethernet, refer to the Setting up Ethernet section below. Note that there is no standardization of MagJack pinouts. You must use the specified MagJack model, see the BOM section for part number or purchase the Ethernet Kit.

Dust Collector/Vacuum Control

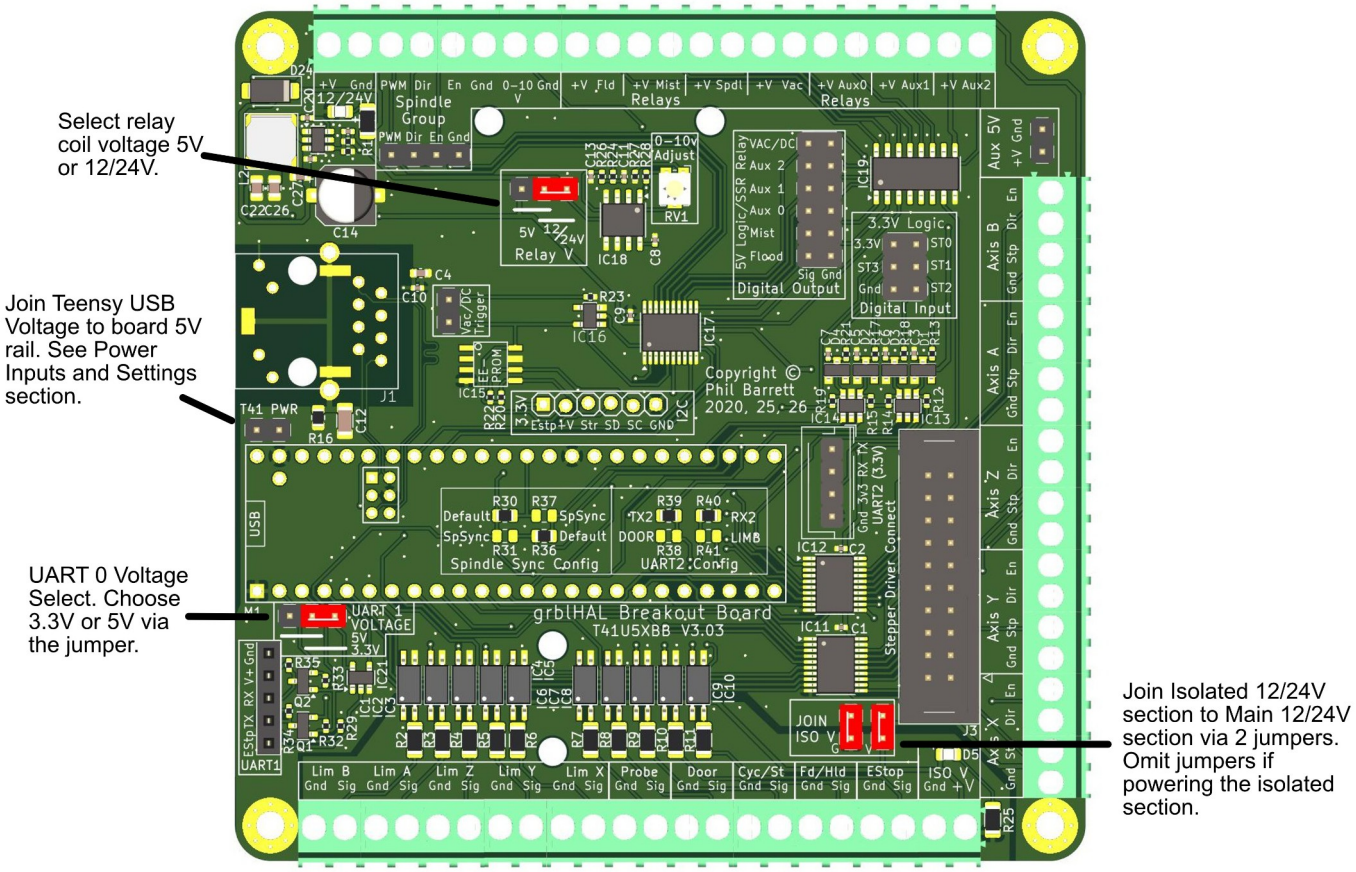


An additional driver is provided for a dust collector relay. It is activated whenever the spindle is turned on (M3). In addition, this relay output can be independently activated by a switch that connects to the 2.54mm 1x2 pin header labeled Vac/DC Trigger. This makes it easy to do clean up with the same Vacuum or Dust Collector used for dust extraction while running your CNC machine.

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Initial Board Setup

The T41U5XBB has a number of jumpers that you will need to set up to make your board ready for the next steps. Note the position of the red shorting plugs (your assembly kit may have a different color). The board shown below is setup for USB connected, single 12/24V PSU configuration. For different configurations, see the Power Inputs and Settings section.



Preparing for Lathe usage

If you are building or upgrading a CNC lathe, you will want to enable spindle sync for threading or electronic lead screw operation. The requires move two zero ohm resistors on the T41U5XBB. This is best done prior to installing the sockets for the Teensy 4.1. See the Board Level Configurations Options section below.

In addition, you will use a different board selection (T41U5XBB Spindle Sync) for building the firmware. See the Web Builder: Spindle Sync section below.

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Board Level Configuration Options

There are several options to modify the behavior of the T41U5XBB board.

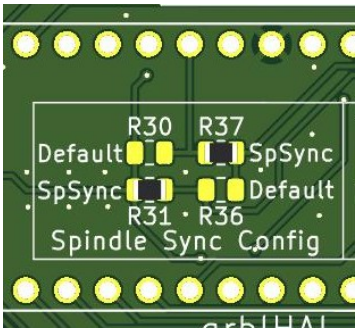
Zero Ohm resistors

Several options are controlled via zero ohm resistors. To move a resistor, use a soldering iron and alternatively heat each pad of the resistor until both side are motlen and push the resistor off the pads. To solder the resistor to a different place, heat on of the destination pads and add a small blob of solder. With tweezers, position the resistor in line with the two pads toward the center, head the solder blob until it melts and slide the resistor in place. The other end of the resistor should be directly in the center of the other pad. Touch the soldering iron to that end and solder it in place.

If you accidentally lose the zero ohm resistor, you can use a small piece of stripped solid 20 to 24 gauge wire. Make a 90 degree bend in the wire so the short piece covers both pads. Holding the wire with needle nose pliers, solder the cut end to the first pad. Then solder the wire to the second pad. Finally, trim the excess wire.

Spindle Sync

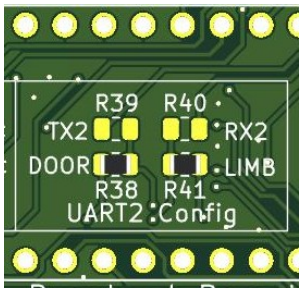
This option enables high speed synchronization signals to support threading for lathe operation. Move the zero ohm resistors R30 and R36 to R31 and R37. The Spindle Sync Config area is underneath the Teensy 4.1. The default configuration does not enable Spindle Sync. See the figure below for the position of the resistors for the Spindle Sync configuration. Alternatively, you can use a short piece of wire in place of the resistor. Look for the Enabling Spindle Sync blog post on grbl.org <insert URL when done>. This is best done prior to soldering in the Teensy sockets.



Spindle Sync requires a different board selection. See the Web Builder: Spindle Sync section below.

UART2, Safety Door and Limit B

UART 2 shares its pins with the Safety Door and Limit B pins. As such only one use is allowed.The UART2 Config area is underneath the Teensy 4.1. The default is for UART2.If you need Safety Door or Limit B, move the zero ohm resistors R39 and R40 to R38 and R41. See the figure below for the position of the resistors for the Spindle Sync configuration. Alternatively, you can use a short piece of wire in place of the resistor.



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Solder Jumpers

The next several features involved the use of solder jumpers. To close a solder jumper

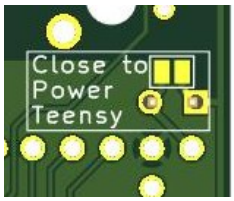
- Use a soldering iron and heat up one of the two pads.
- Apply a generous amount of solder.
- Repeat the process for the other pad.
- Work the soldering iron back and forth between the two pads, pushing the molten solder towards the middle. Add solder until the two blobs bridge together.
- Allow the solder to cool and inspect to make sure they are joined.

To remove a solder jumper, use solder wick.

- Apply solder flux to the solder bridge and solder wick.
- Heat the bridge with the iron until it melts.
- Press the wick onto the hot bridge with the soldering iron until you see the solder flow into the wick and remove the solder wick before the solder solidifies.
- Repeat the process with a fresh section of the solder wick until the solder bridge is removed.

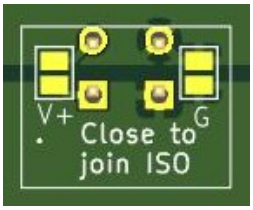
T41 Power Solder Jumper

This solder jumper on the bottom of the board is the same as using a shorting plug on the T41 PWR header. See the Power Inputs and Settings section for more details on this jumper. Do not close this jumper without reading that section.



Join ISO V Solder Jumpers

These two solder jumpers on the bottom the board are the same as using shorting plugs on the JOIN ISO V headers. See the Power Inputs and Settings section for more details on this jumper. Do not close this jumper without reading that section.



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Reverse Polarity Diode Bypass

A diode is used to prevent board damage from reversing the polarity of the 12/24V input wires. While it is unlikely to cause problems, the diode causes a small voltage drop. A low 12V input (11.5V or so) may interfere with generating the 0-10V spindle speed control signal. The best solution to this problem is to adjust the 12V PSU to have at least 12.0 V output. In the event that is not possible, you can remove the diode by closing the solder jumper in the RP BYPASS section on the bottom of the board.



Warning: bypassing the RP diode removes the reverse polarity protection. Reversing the supply voltage inputs will cause damage to the board. RP bypass should only be done as an absolutely last option.

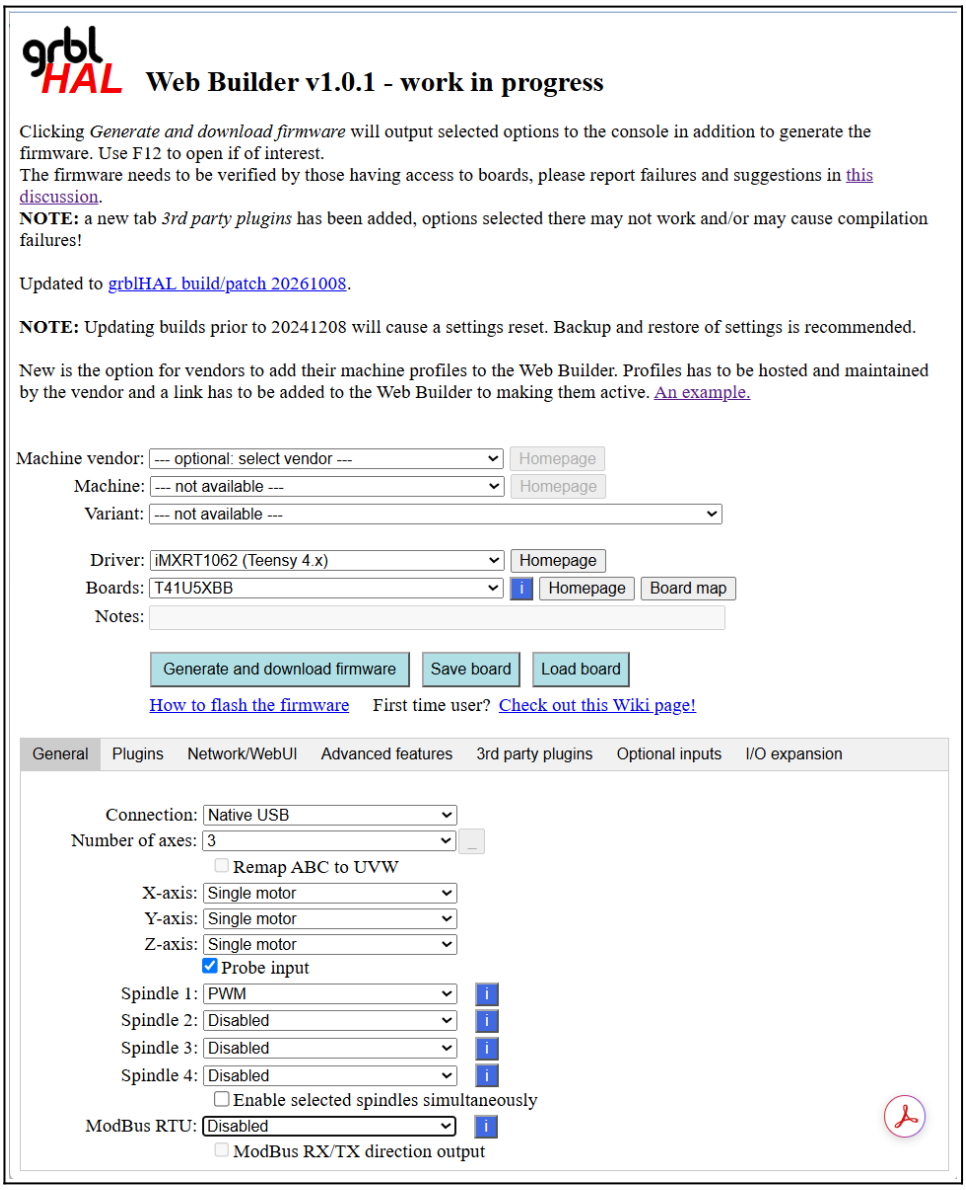
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Building and Loading Firmware.

The grblHAL Web Builder is the primary tool for building firmware for the T41U5XBB. It is available at https://svn.io-engineering.com:8443/.

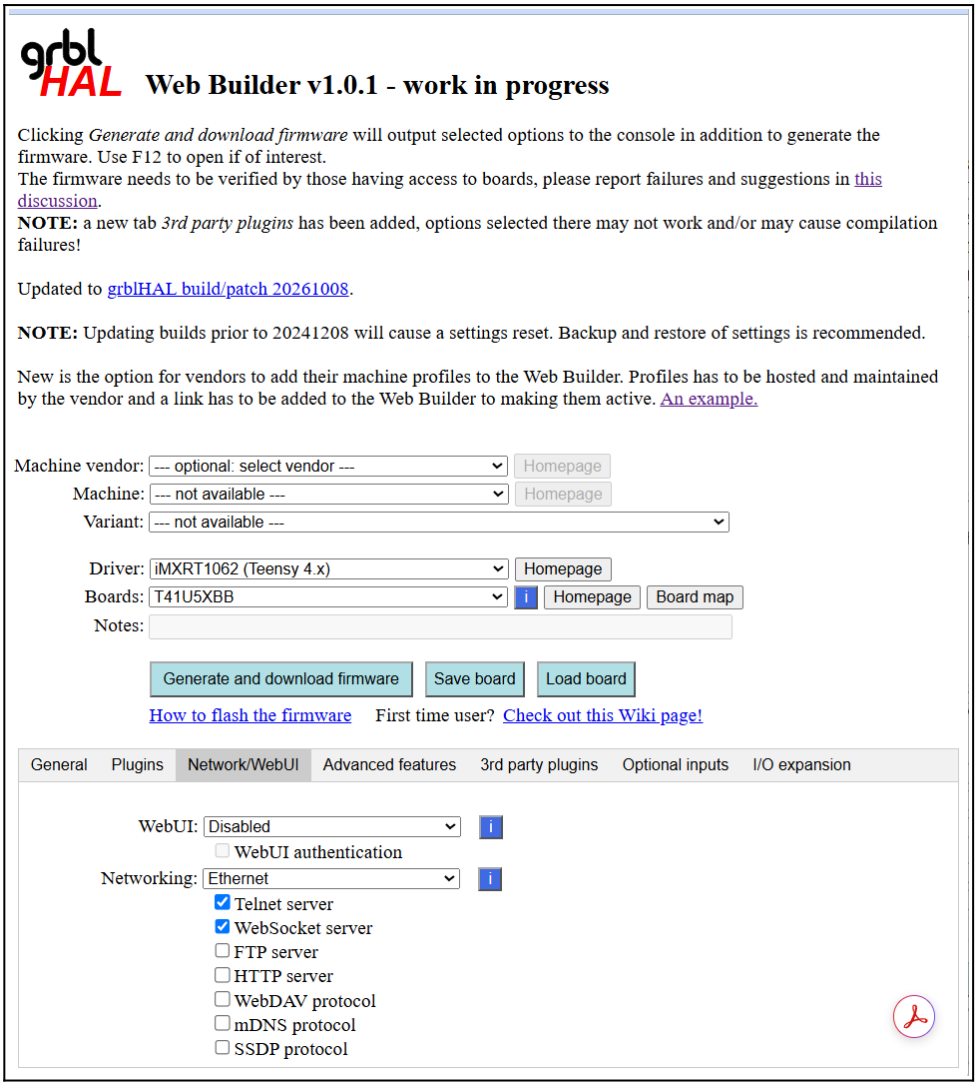
On the main panel select iMXRT 1062 (Teensy 4.1) as the Driver and T41U5XBB as the Board. In the General panel, select the Number of axes. If you have a dual motor moving gantry, select that axis as Ganged or Auto-squared. The second motor will be A Axis if you selected 3 Axes or Axis B is you selected 4. If you have installed the Ethernet kit, select Ethernet in the Network/WebUI panel and check Telnet server and WebSocket server options. Press Generate and Download Firmware button and save the file for loading. Be sure to save the configuration by clicking on the Save Board button and name the file something memorable. This will allow to you to later make changes without having to remember what you selected.

Web Builder: Main screen with General panel



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Web Builder: Network panel



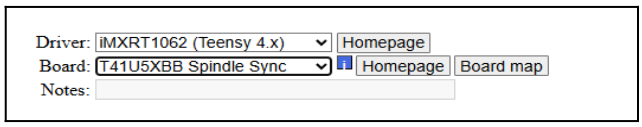
Web Builder: Other features

There are numerous features available in the Web Builder, far too many to cover in this document. We recommend looking at each panel to see the additional features. Most have an i icon that links to information about the feature.

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Web Builder: Spindle Sync

If you are building a CNC lathe and wish to use spindle sync, you will need to use a different board definition – T41U5XBB Spindle Sync. All other options are selected as suggested above. You will also need to make a board level configuration change to support this. See Spindle Sync in the PCB Level Configuration section.



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Loading Firmware

You will use the teensy.exe application available from pjrc.com. Start with this page - https://www.pjrc.com/teensy/loader.html and follow the links for your specific operating system. In general, you run Teensy.exe, load the specific firmware file (generated by the grblHAL Web Builder) and then press the button on the Teensy 4.1. You should see erasing and programming messages in the teensy.exe application.

Building from source code

Building grblHAL from source code is well beyond the scope of this document. You can download the source code at https://github.com/terjeio/grblHAL To build it, follow the instructions for Compiling GrblHAL in the grblHAL Wiki at https://github.com/terjeio/grblHAL/wiki/Compiling-GrblHAL. The grblHAL core Discussions section (https://github.com/grblHAL/core/discussions) is a good place to ask questions and get help.

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Running grblHAL on your machine

Once you have finished assembly, your grblHAL Teensy 4.1 based Motion Controller is ready to be tried out. Attach it to a PC via a USB cable and run a compatible GRBL sender application on the PC. We recommend ioSender and use it in our examples: https://github.com/terjeio/ioSender. Check for the most recent release. An overview of ioSender is available here: https://www.grbl.org/single-post/one-sender-to-rule-them-all.

Next, you will need to change the settings to reflect your hardware. Actual grbl setting values are also shown. Follow these steps:

1. If using, hook up your control buttons: Feed Hold, Cycle Start and Reset/EStop.
2. Start ioSender and open the **Settings: Grbl** panel. You will see a list of parameters that control grblHAL.
3. Set the “Invert Step Enable” mask. Check all boxes. $4=7 for a 3 Axis machine. $4=15 for a 4 Axis machine and $4=31 for a 5 Axis machine
4. Set “Invert Control Pin” mask. Invert Feed Hold, Cycle Start, Reset and Safety Door. $14 = 70.
5. Save these parameter and return to the main tab (**Grbl** panel, you should see the DRO section).
6. Press the Red Reset button if an ALARM state is showing. Press Unlock.
7. Open the **Jog** panel and click on any of the jog buttons. The DRO should show movement. If you have motors hooked up, they should show movement.
8. If you have any control buttons attached, test them. It should show activity. Note the associated “LED” in the G Code Sender application when you press a button. If nothing happens, you should verify your connections.
9. Close the **Jog** panel, in the Spindle section, enter 1000 in the Spindle RPM box and click on the CW “radio” button. The LED on the Teensy should light up. Click on off and the LED should go off.
10. If everything works to your satisfaction, proceed to full testing of your machine.

You can also refer to the grblHAL wiki's First Run Grbl Settings section: https://github.com/grblHAL/core/wiki/First-Run-Grbl-Settings

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Some initial Grbl Settings

Grbl settings are dependent on your hardware so you might have to experiment a bit. Here are some key settings that will help you get going quickly. You should spend some time studying all the Grbl settings. We suggest you start conservatively and tune your machine to higher performance one parameter at a time.ependent on your hardware so you might have to experiment a bit. Here are some key settings that will help you get going faster.

$ID	Name/meaning	Suggested Value	Comments
0	Step Pulse time in uS	3	Default is 5 but 3 will work well. Using the default will cause problems for step rates above 140 kHz. Minimum is 2 uS.
4	Step Enable invert bit mask, one per axis	7 (3 axis), 15 (4 axis) 31 (5 axis)	Bitmap field. A 1 bit means invert. A 3 axis machine would have 7 as the value, a 4 axis machine would have 15 and a 5 axis would have 31. Bit 0 is X, 1 is Y, 2 is Z, A is 3 and B is 4.
5	Invert Limit Input bit mask, one per axis	7 (3 axis) 15 (4 axis) 31 (5 axis)	Bitmap field. Set each bit to 1 to use N.O. Switches/sensors as limit inputs. For N.C. Switches/sensors, set each bit to 0.
14	Control pin inversion bit mask, one per pin.	70 - invert Feed Hold, Cycle Start and Estop	A 1 bit means invert. Reset is bit 0, Feed Hold – 1, Cycle Start – 2, Safety Door – 3, Estop is 6. This assumes you are using N.O. switches for input. N.C. switches will need a 0 bit.
30	Max spindle speed in RPM	1000 or VFD max.	Default is 1000. If you have a VFD, you will need to set this to the spindle max. In G-Code, the Sxxx parameter controls speed and to get full speed, you will need to have xxx match this parameter (S1000, in the case of the default).
100, 101, 102, 103, 104	Steps/mm. One for each Axis. 100 – X, 101 – Y, and so on.	varies	Set these based on your hardware – lead screw/belt ratios, microstepping. The Grbl defaults are likely wrong for your machine but you will still get movement. On a per axis basis, determine the number of steps (including microsteps) it will take to make one full rotation of the lead screw. Determine how many mm that axis will move with one full rotation of the lead screw, pinion gear or belt driver pulley. Divide the first number by the second to get the value you will use. See Calibrating grblHAL CNC system movement for fine tuning these settings.
110, 111, 112, 113, 114	Max Rate 110 – X 111 – Y and so on.	2500 mm/Min	This is a reasonable starting point for a hobby machine. Higher performance machines can easily support 5000 or higher. If your machine has missed steps or chatters, dial it back.
120, 121, 122, 123, 124	Acceleration in mm/Sec^2 120 – X 121 – Y and so on.	500 mm/Sec^2	This is a reasonable starting point for a hobby machine. High performance stepper motors and servos can support significantly higher acceleration. If your machine has missed steps or chatters, dial it back.

Note, if you don't attach any switches, the system may come up in the alarm state before you change any settings. This is normal. If you are using the Safety Door Alarm, you may need to invert Safety Door setting. Similarly, you can invert the Halt input. However, we highly recommended that you do not disable the Halt function and that you install a dedicated EStop button connect to Halt in a prominent location on your CNC machine.

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You should spend some time studying all the GRBL settings. We suggest you start conservatively and tune your machine to higher performance one parameter at a time.

Compatible G-Code Senders

The vast majority of GCode Senders work with grblHAL. If your preferred sender is having problems, you may need to use a version of grblHAL built with compatibility settings. The grblHAL Web Builder allows you to select compatibility level when building firmware.

Setting up Ethernet.

If you wish to use Ethernet to connect to your breakout board, you will need to solder the network interface connector MagJack, 2x3 2mm header and 2x3 2mm socket in place. See the Assembly section above. It is available in the Ethernet kit. See the Bill of Material section for the part numbers if you want to source the Ethernet components yourself. Once you have soldered them on the PCB, you will need to load a build of grblHAL that has been configured to use networking. You will use the USB connection to do that.

You will next need to load an Ethernet build of the firmware, available via the grblHAL Web Builder (https://svn.io-engineering.com:8443/). Optionally, you can build a version from the source code at the grblHAL Github site.

To set up Ethernet, refer to this blog post for instructions: https://www.grbl.org/single-post/networking-the-t41u5xbb

Direct PC Connection via Ethernet

Many CNC users like to use a direct Ethernet connection between the PC and CNC machine. The advantage of this is the CNC machine is not visible from your network and is thus safe from intrusion and interference. Because grblHAL is not a secure system, this is the only way to network secure your CNC machine. Refer to this blog post for instructions: https://www.grbl.org/single-post/ethernet-without-routers-for-grblhal

Note that modern network switches prevent most interference and are quite inexpensive.

0-10V output setup

The 0-10V circuit works by using an amplifier to filter a 5V PWM signal and boost it to 10V. As such it needs 12V or higher input which the user must provide via the 12/24V screw terminals. A second order filter with an aggressive low pass cut off frequency is used to provide a stable signal for the spindle controller.

Because source voltage and input levels can vary, the amplifier must be adjusted. You can do this by turning on your spindle in GCode (via Sxxxx) to full speed and adjusting RV1 until the voltage on the 0-10V terminal reads 10V. Here is a step-by-step method you can use:

- Disconnect your spindle or VFD.
- In the **Settings: Grbl** tab, set $30, **Maximum Spindle Speed**, to 1000.
- Send the following command to grblHAL: **M3 S1000.** This turns on the spindle and causes the speed to be set to 100%. The LED on the Teensy 4.1 should light up.

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- Measure the 0-10V output and G terminals.
- With a screwdriver, adjust RV1 (labeled “0-10V Adjust”) until the voltage is as close to 10V as you can get. Anything in the range 9.990V to 10.010V is acceptable.
- Send the following command to grblHAL: M5. This turns off the spindle.
- Reconnect your spindle or VFD and test.
- You may want to change the max spindle speed setting, $30, to something more appropriate for your spindle/VFD.

You may want to repeat the above process with the spindle connected and running to fine tune the maximum speed.

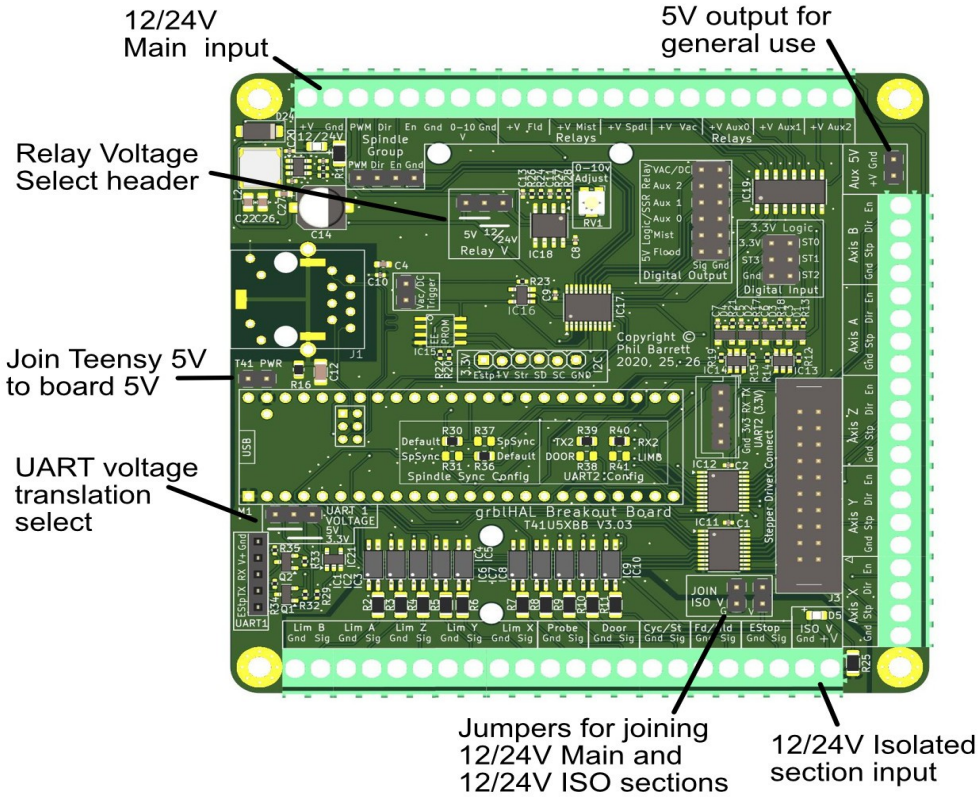
If you have a VFD or spindle controller with a speed readout attached, you can use the readout to tune the 0-10V output:

- Set the Maximum Spindle Speed setting ($30) to your spindle’s maximum speed, typically 24000 or 18000.
- Send the following command to grblHAL: **M3 S<spindle maximum speed>.** This turns on the spindle and causes the speed to be set to 100%. The LED on the Teensy 4.1 should light up.
- With a screwdriver, adjust RV1 (labeled “0-10V Adjust”) until the readout shows the maximum rated speed**.** You may want to back off the speed to find the lowest setting that reads the maximum rated spindle speed.
- Turn off the spindle by sending an **M5** command to grblHAL.

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Power Inputs and Settings

This diagram shows the various power related headers and inputs.



Isolated vs Main 12V/24V

The T41U5XBB has an isolated 12V/24V section for limit switches and input controls. This is not connected electrically to the rest of the board which uses the 12V/24V main supply. Using a separate 12V or 24V PSU for the Iso section is the recommended configuration for best EMI immunity. The Join 12V/24V jumpers allow a single PSU to be used if you wish to economize. In many cases, the opto-isolators provide sufficient isolation without a separate 12V or 24V PSU.

The T41U5XBB requires 12V or 24V input. It will draw up to approximately 200 mA, though any directly driven relay coils or other devices will add to that.

Warning

- Don’t use a PSU that may exceed the board’s rated maximum input of 24V. Unregulated wall adapters (aka “wall warts”) can go quite high. These can damage the T41U5XBB. Use a regulated PSU that outputs 12V or 24V even with no load.
- Don’t reverse polarity if you have bypassed Reverse Polarity protection. Connect plus (+) on the PSU to (+) on the T41U5XBB board. Connect minus (-) (or Gnd) on the PSU to Gnd on the T41U5XBB board.
- Don’t “hot plug” the 12V/24V source. This means plugging an already powered PSU into the T41U5XBB. It may cause a voltage surge which can lead to circuit damage. Best practice is to connect the PSU before powering it on.

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USB Power and the Teensy 4.1

The Teensy 4.1 by default is powered from the USB connection but that presents some complications. Especially if you want to use Ethernet and completely remove the USB cable except for loading firmware. The T41 PWR header allows joining the onboard 5V regulator to the Teensy 4.1 5V pin. Best practice is to use the onboard 5V regulator to also power the Teensy 4.1 (ie NOT the USB connection). In general, you want to avoid powering the Teensy 4.1 via USB and have board generated 5V connected to the Teensy (ie, with the T41 PWR jumper on). You have several options. Option 3 is the preferred solution.

1) Power the Teensy via USB, T41U5XBB via onboard regulator

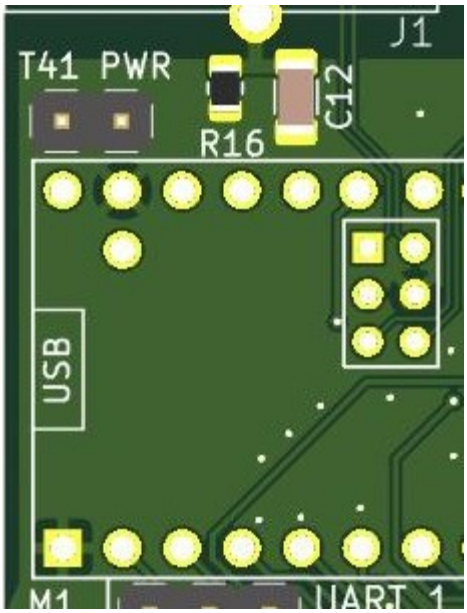
Do not jumper the T41 PWM header (or close it via solder jumper on the bottom). You will have to use the USB connection even if you use Ethernet to connect.

2) Remove USB, use Ethernet and everything via the on board regulator

Jumper (or close via the solder jumper) the T41 PWR header. This requires that when installing firmware (which uses USB), you either use a data only USB cable or remove the T41 PWR jumper when installing firmware and configuring networking.

3) Modify the Teensy 4.1 to disconnect USB Power.

You can cut a trace on the Teensy 4.1 to disconnect it from USB power. See the blog post Isolating the Teensy 4.1 from USB Voltage for an in depth discussion of this option. You will also need to install the T41 PWR shorting plug (or close via solder jumper). This is the preferred option as it always avoids the on board regulator and USB power fighting with each other. It prevents accidental power supply conflict.



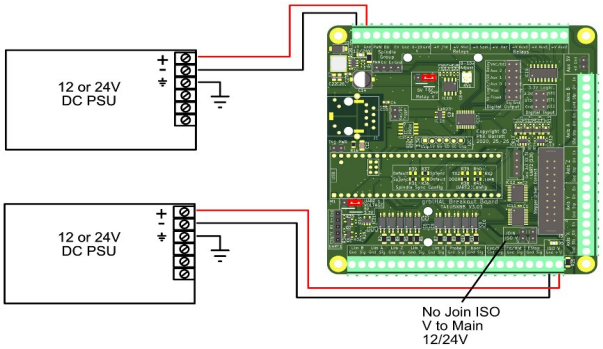
T41 PWR Header

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Three Ways to Power the T41U5XBB Board

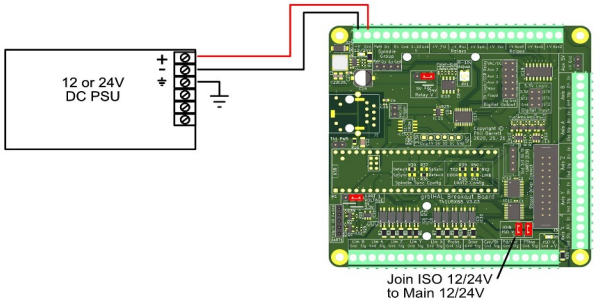
1) Dual Power Supply Units, fully isolated (recommended).

This provides separate power for the relay and spindle sections as well as isolated power for the limit sensors/switches. The digital section – the Teensy itself and all the digital logic on the T41U5XBB – can either be powered by the on-board 5V regulator fed from the 12V or 24V main input or from the USB connection. See the USB Power and the Teensy 4.1 section above. A 1 Amp PSU is acceptable for either supply. If you have other 12V or 24V items to power, use the main section PSU and size accordingly to the additional load. Note that the PSUs do not have have the same voltage in this case.



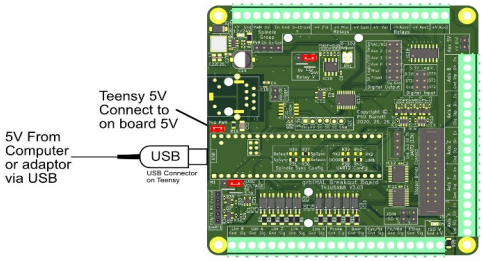
2) Single PSU.

For budget conscious builders, there is the option to use a single PSU. The two 2-pin headers near the 12V/24V Iso input can be jumpered to join the two 12V/24V sections together and thus have one 12V or 24V PSU power the entire board. This does not provide true isolation for the limit, probe and control input section but may be acceptable, especially if good EMI practices are followed in your CNC system. The opto-isolators provide some EMI protection even without a fully isolated PSU. See Electromagnetic Interference below. The digital section – the Teensy itself and all the digital logic on the T41U5XBB – can either be powered by the on-board 5V regulator fed from the 12V or 24V main input or from the USB connection. See the USB Power and the Teensy 4.1 section above.



3) Logic Powered from USB

For initial testing outside the electronics case without a 12 or 24V PSU, the core digital section of the T41U5XBB can be powered via USB power. This is acceptable if you don't power any relays or other devices. However, the Limit, Probe and Input sections require 12V or 24V power to function. If you don't supply 12V or 24V to the T41U5XBB, grblHAL may boot up in an error state. See First Run Grbl Settings in the wiki for how to deal with that. Note the position of the shorting plugs in the image to the right.



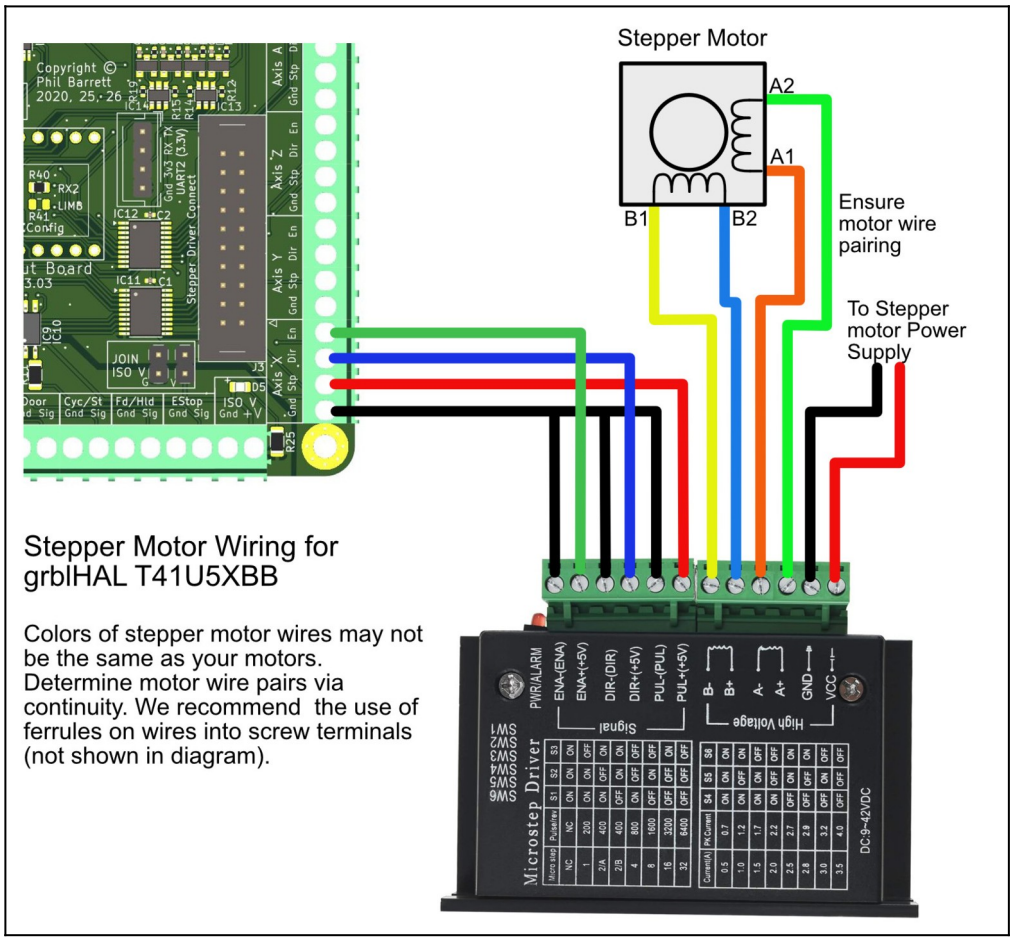
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External Device Connections

Using Stepper and Servo Drivers with the T41U5XBB

Using stepper motors with the T41U5XBB is fairly simple. Typically, a Stepper Motor Driver is used to turn the T41U5XBB's Step and Direction signals into voltage and current that cause the desired step motions of the stepper motor. Servo drivers work similarly, so for simplicity we will just talk about steppers here but the following applies to servos as well.

Most stepper drivers use opto-isolators for input so you will see a plus (+) and minus (-) terminal for each signal. In our case, we will connect the minus (-) inputs to ground on the T41U5XBB board and the plus inputs (+) to the named signals (Step, Direction and Enable) on the T41U5XBB board. Typically, the Step terminals on drivers are labeled PUL (pulse), the Direction terminals are labeled DIR (direction) and the Enable terminals are labeled ENA (enable). Other stepper drivers may have variations on these names.



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Active Low Stepper Drivers

While most modern stepper drivers expose positive and negative signal connections, some (especially older) drivers only expose the negative connections. The T41U5XBB can drive those easily. The positive input side of the driver is connected to a +5V output (use the +5V Aux header on the T41U5XBB). Step, direction and enable signals from the T41U5XBB connect to negative signal inputs on the driver. You may also need to invert Step, Direction and Enable signals via the $2, $3 and $4 Grbl settings. See this blog post for more details. It uses an older board (PicoCNC) but is still applicable.

Using Servos with the T41U5XBB

The connections are similar to those in the diagram above. In addition, servos also have an alarm output, often labeled ER, ALM or similar. This signal indicates when the servo has failed to operate as directed, often due to lost steps. The T41U5XBB has several inputs that are compatible with servo alarm signals: Estop, Door and Limit inputs. This requires open-collector outputs (common on most servo drivers).

The recommended grblHAL input pin for the servo alarm signal is the EStop/Halt signal. You wire the Alarm output of the servo controller to the Estop
input screw terminal. When an alarm is asserted, the controller will enter the halt state. This will cause your G-code program to stop and the spindle to power down. The EStop/Halt input pin must be inverted and the physical EStop switch must be normally open (N.O.).

It is also possible to connect servo alarm output for a given Axis to the associated limit input. It can be connected in parallel with an N.O. NPN sensor on the associated axis limit input. PNP and N.C. sensors are not supported. You will need to configure your Grbl Settings to use hard limits and have correct inversion on the limit input.

Many servo controllers allow configuration of the alarm output. These go by different names “sinking”, open drain or open-collector are common. Select that when connecting to the various inputs of the T41U5XBB.

Note: If you switch the Alarm input to Door, you will need to rebuild the firmware with Safety Door enabled and change the board level configuration (Door is not a default configuration). In the grblHAL Web Builder it is not enabled by default. The Safety Door check box is on the Optional inputs panel.

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Connecting to a Spindle

There are several different ways you can connect the T41U5XBB board to control a spindle:

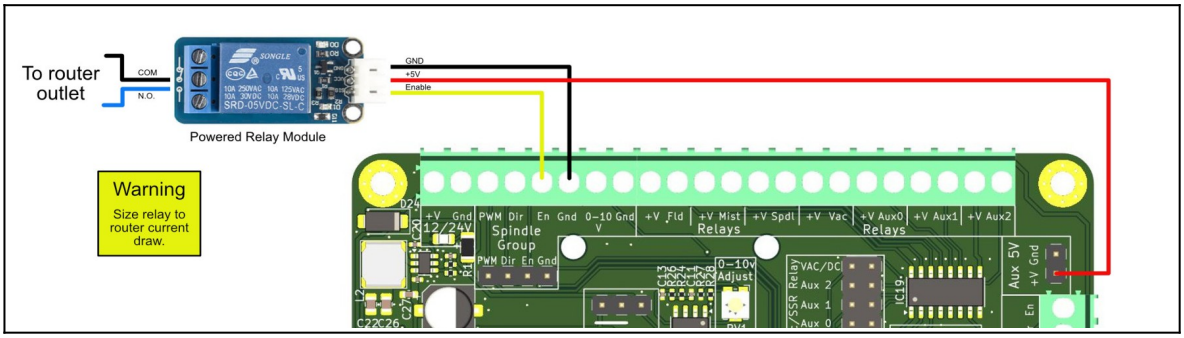
- On/Off control via Spindle Enable (En) or Relay.
- PWM control of a speed controller or VFD.
- 0-10V control of a VFD.
- Modbus/RS485.

On/Off control

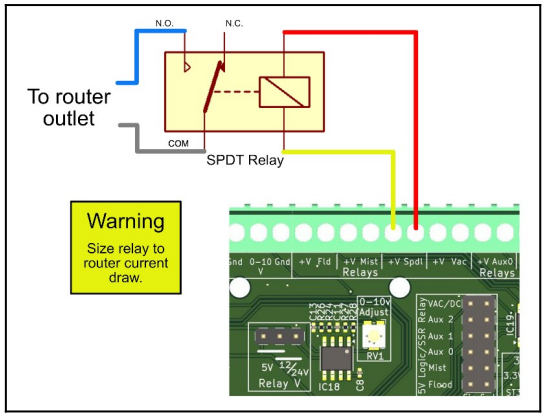
There are two places to control spindle On/Off: the En screw terminal in the spindle group (a 5V logic level output) or the SPDL EN screw terminal (an open-collector output for a relay coil). Either of these can be used to control a relay to turn a spindle on or off. This is how you would use a router as the spindle.

Cautionary Notes:

- If you are unfamiliar with AC voltage, consult a professional electrician before proceeding.
- Be conservative in selecting a relay with the right AC current rating. Most routers have a surge current that can be significant. Note also that inexpensive relays are often optimistically rated. Again, if unfamiliar with AC relays, consult a professional electrician.
- Only use connectors, plugs and outlets that are approved for line voltage and high current in your jurisdiction. Consult a professional electrician if you are not familiar with line voltage.
- Read the Using Relay and Digital Outputs section for information about relays with the T41U5XBB.



On/Off control via a powered relay module

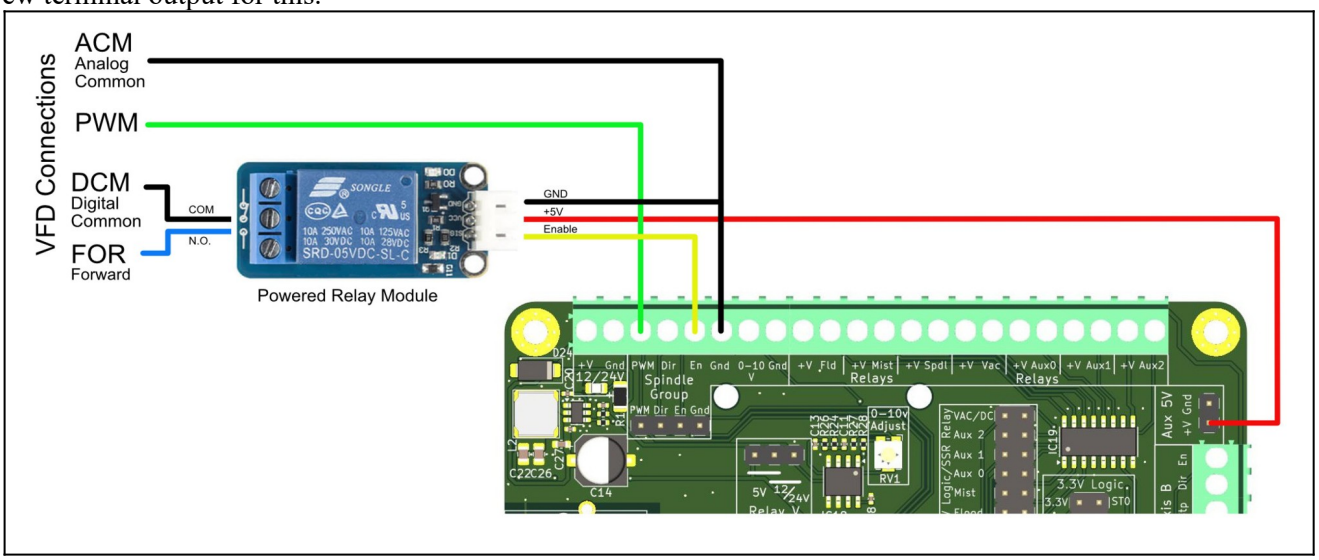


On/Off control via driving a relay coil

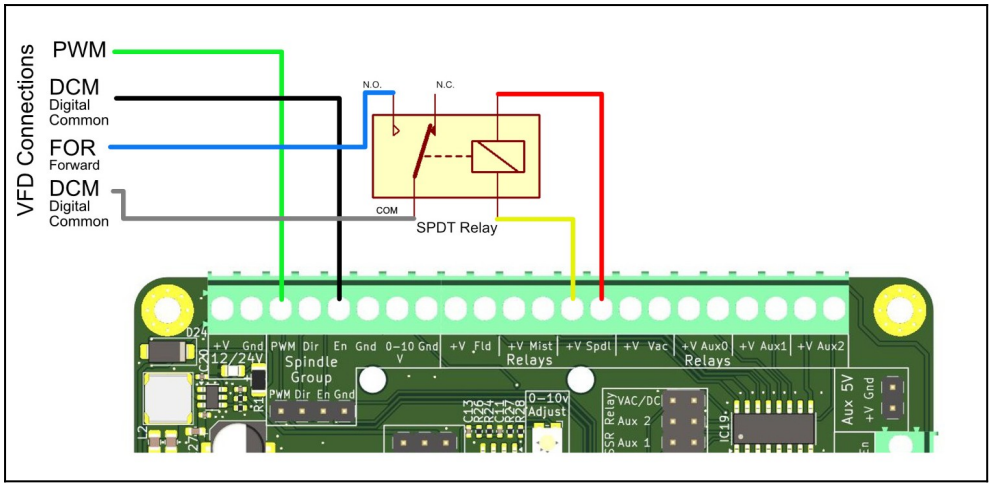
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VFD Control via PWM

Some spindle controllers take 5V PWM control. This is common in DC motor controllers. You can use the PWM screw terminal output for this.



PWM control via a powered relay module

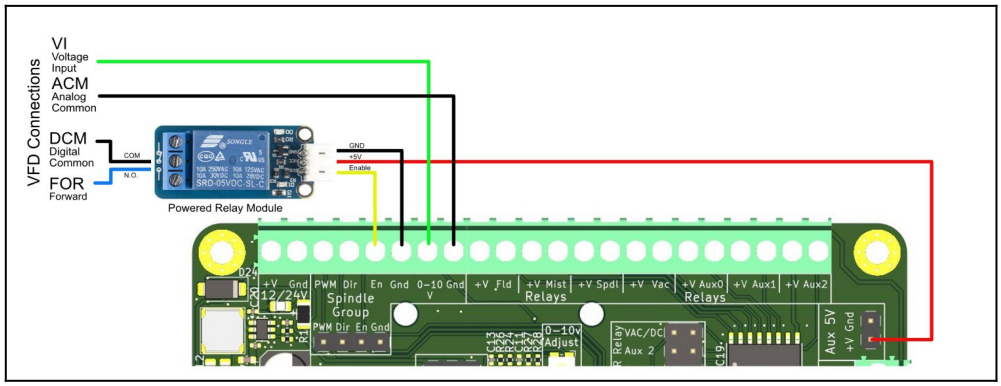


PWM control via driving a relay coil
Make sure you select the correct relay voltage.

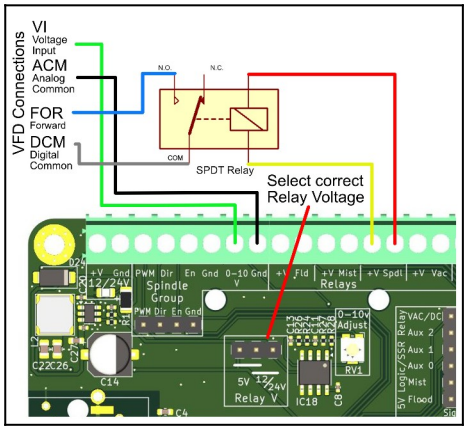
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VFD Control via 0-10V Output

VFDs may be controlled via 0-10V output. A typical connection is shown below. The VFD must be configured to take an external voltage for control and set to 0-10V. Some VFDs will also take PWM control. The actual configuration steps for a VFD are beyond the scope of this manual. Consult your VFD's user manual.



0-10V control via a powered relay module



0-10V control via driving a relay coil
Make sure you select the correct relay voltage.

Modbus/RS485

This is not directly supported on the T41U5XBB board, but you may use a serial-to-RS485 (Modbus) adapter with either UART header.

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Using Relay and Digital Outputs with the T41U5XBB

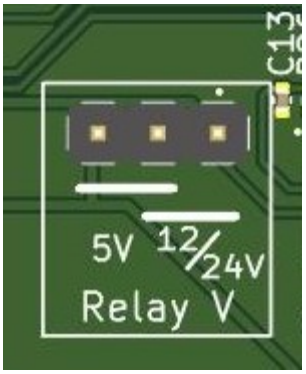
There are several ways to use these outputs on the T41U5XBB board:

- To directly drive external 5V, 12V or 24V relay coils.
- Drive solid state relays (SSR).
- Drive powered relay boards (PRB) with 5V control inputs.
- Drive 5V logic (TTL) devices.
- As a general purpose open-collector (OC) driver.

A note on using relays and other devices to switch 110/220 VAC: Please be aware of the risks associated with using relays to control high voltage devices like motors. If you are not familiar with high voltage, we recommend you consult a professional electrician for help in selecting the right devices and proper wiring techniques.

Relay Voltage

Relay Coil Voltage source is selectable between 5V and 12/24V provided via an external power supply. Use a 2.54mm shorting plug (provided in the basic assembly kit) as indicated on the PCB to select the desired voltage. The selected voltage will be present on the +V screw terminals.



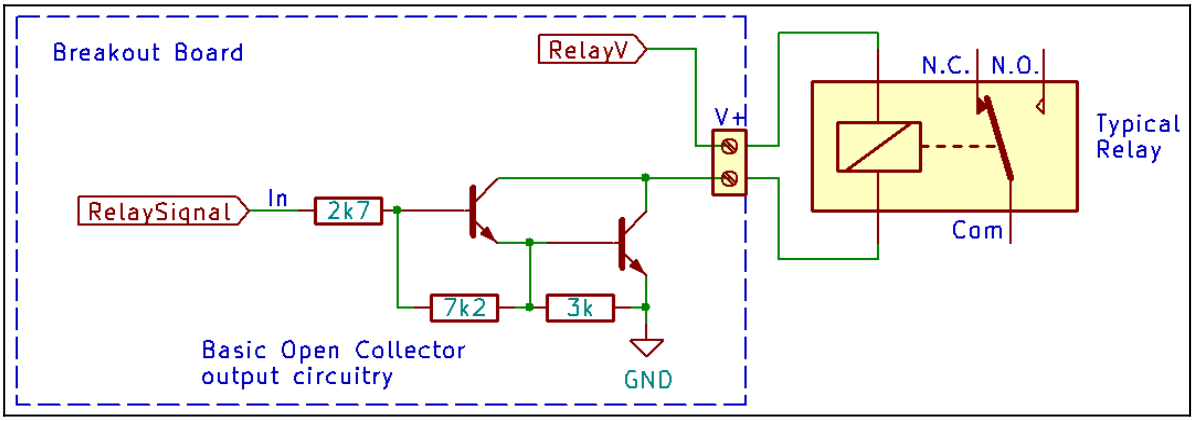
Relay coil voltage up to 24V is permissible and is applied at the input terminal labeled 12/24V.

Note: it is recommended to not use the stepper motor supply for the relay supply. It could be a source of EMI and cause erratic operation of the system.

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Directly Driving External Relay Coils

The relay screw terminal outputs of the T41U5XBB board are used to drive relay coils. Automotive and DIN Rail mountable relays are commonly available. The relay coil is wired across the two terminals as shown below. Relay coil current up to 100 mA is supported. The 12V/24V relay section PSU supplies the current for relay coils. Add up the total relay coil current draw and add 200 mA. Make sure that your PSU is capable of delivering that amount of current.



Driving Solid State Relays

SSRs are driven from the Digital Output section pin headers. The output voltage level is +5V, which is compatible with most SSRs, but verify that yours is compatible before proceeding. DIN rail mountable SSRs are readily available.

Driving Powered Relay Boards

Powered Relay Boards are driven from the Digital Output section pin headers. The output voltage level is +5V. Make sure that your PRB is compatible. PRB modules are readily available. Be aware that some cheaper modules use optimistically specified relays. We recommend derating relay contact ratings by at least 20% for safety.

Driving 5V Logic Devices

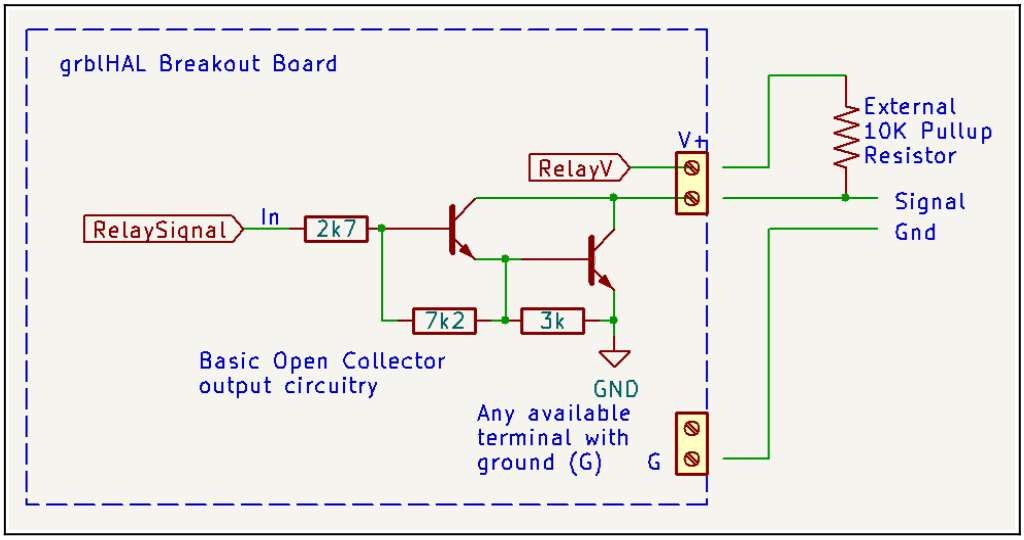
The Digital Output section pin headers are usable for general 5V logic signaling. For Aux 0–Aux 2, use the M62-M65 commands to turn them on or off. These commands can be issued from a G-code program, from a macro in your G-code sender, or typed directly into the sender's console.

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General Purpose Open-Collector Driver

The open-collector outputs of the relay screw terminals are available for general use. The current is limited to a sum total of 500 mA through all terminals at a 100% duty cycle. Any single terminal is rated for 100 mA. The V+ terminals supply either +5V or +12V/24V, determined by the jumper on the Relay Voltage header. A common ground may be needed. Any unused ground terminal on the relay or stepper output section of the T41U5XBB can be used. Don't use a ground pin from the Limit or Control input sections.

Use an external through-hole resistor if you need a pull-up resistor for your application,.



Open-collector circuit on the T41U5XBB

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Power and EMI Considerations

Clean, EMI-free power is important for reliable operation of the grblHAL motion controller system. In designing the T41U5XBB, we used good practices to avoid EMI but further precautions are recommended. Power is often a source of EMI and you should take additional steps to ensure that these sources are managed. Finding the source of interference later can be difficult – it is good practice to design your system with avoiding it in mind. General recommendations:

- Do not power the board directly from the Stepper Motor power supply. It can be a significant source of EMI.
- Use shielded wires for signals. Makes sure the shields are grounded in a star configuration at the controller box or other common ground.
- Use ferrite filters on all wires entering or exiting the T41U5XBB PCB.
- Avoid routing wires near EMI sources such as a VFD, Stepper drivers, or power supplies.
- Avoid co-routing with wires that are connected to those devices as well.
- If you are connecting to the Teensy 4.1 via a USB cable, use one with a ferrite filter on it or add a clamp-on one close to the Teensy end of the cable.

If you are using Ethernet, make sure the 5V SRC header is jumpered to OB (on board). You may disconnect the USB port.

We recommend that all stranded wires have ferrules crimped on the end that goes into a screw terminal. This is a good practice in that it ensures the best possible contact and helps to avoid stray “cat whisker” strands of wire that could cause errors. 24 Ga ferrules are recommended for use with the 3.5mm screw terminals. Slightly flattening the ferrule tip makes it easier to insert.



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Appendices

Appendix 1: Errata

<nothing yet>

Appendix 2: T41U5XBB Revision History

V2.05 Pre-Release version.
V2.07 Initial Version.
V2.09 Minor layout and silk screened label updates.
V2.10 Moved trimmer potentiometer from kit to base board. Update silk screened labels
V2.11 Silk screened label updates
V2.20 Upgraded opto-isolators. Minor layout changes. Improved mounting hole clearance. Added PCB level configuration of Spindle Sync. Silk screened text/label updates.
V2.21 Corrected Lim A and Lim B labels.
V3.01 Not released.
V3.02 Not released. Major update. Larger size PCB. On board generation of 5V. Aux 5V output. External 12 or 24V input. 12 or 24V isolated limit/probe/control input section, Section joinable to main board 12 or24V. Support for UART 2. Spindle Sync configuration via 0 ohm “resistors”, 20 pin Stepper Driver Connect header.
V3.03 Release version. Moved T41 PWR jumper. Opto resistor changes. Numerous silk screen changes.

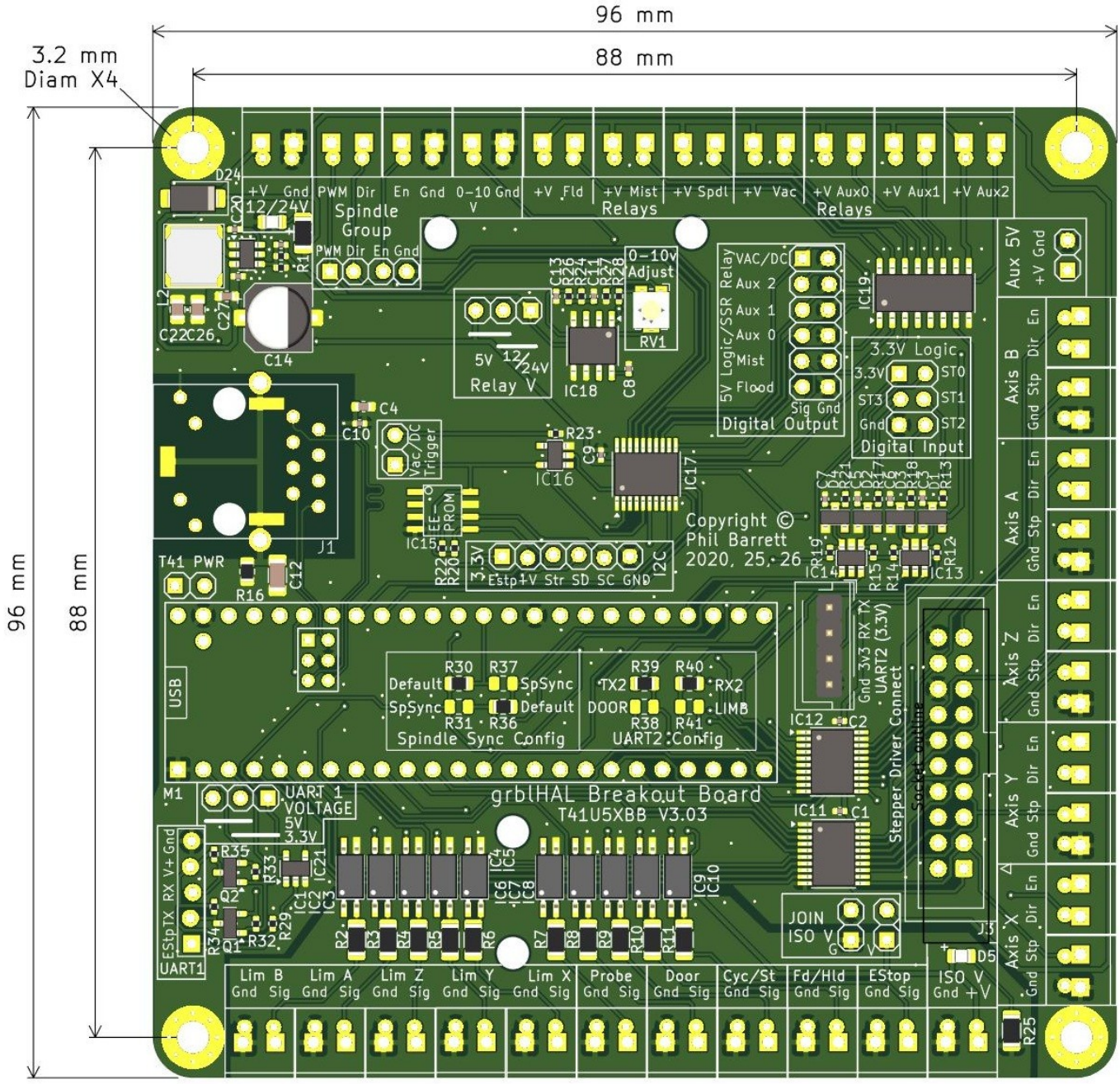
Appendix 2: Schematics

Refer to the T41U5XBB V3 github repository for current version schematics.

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Appendix 3: Board and mounting hole dimensions

V3.03



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Appendix 4: Bill of Materials

Assembly Kit

Item	Number	Source
2 pin 3.5 mm Screw terminal.	32	Readily available on ebay/amazon/aliexpress/...
1x2 2.54 mm pin header	5	Readily available on ebay/amazon/aliexpress/... Snap off of longer strips. [1]
1x3 2.54 mm pin header	1	Readily available on ebay/amazon/aliexpress/... Snap off of longer strips. [1]
1x4 2.54 mm pin header	2	Readily available on ebay/amazon/aliexpress/... Snap off of longer strips. [1]
1x5 2.54 mm pin header	1	Readily available on ebay/amazon/aliexpress/... Snap off of longer strips. [1]
1x6 2.54 mm pin header	1	Readily available on ebay/amazon/aliexpress/... Snap off of longer strips. [1]
2x3 2.54 mm pin header	1	Readily available on ebay/amazon/aliexpress/... Snap off of longer strips. [1]
2x6 2.54 mm pin header	1	Readily available on ebay/amazon/aliexpress/... Snap off of longer strips. [1]
1x24 2.54 mm pin header	2	For Teensy 4.1. If you purchase a Teensy with pins, omit these items. Readily available on ebay/amazon/aliexpress/... Snap off of longer strips. [1]
1x24 2.54 mm socket	2	Readily available on ebay/amazon/aliexpress/... [1][2] Higher Quality: Mouser – 801-87-024-10-001101, Digikey - 1212-1209-ND
Teensy 4.1	1	Not part of basic kit. Available from sparkfun.com.

Ethernet Kit

MagJack	1	Whiznet RB1-125BAG1A Mouser: 950-RB1-125BAG1A Digikey:1278-1011-ND. Do not substitute.
2x3 2 mm Header	1	Digikey:MTMM-103-10-T-D-355-ND, Mouser: 200-MTMM10310TD355 (drop ship from Samtec)[4]
2x3 2 mm Socket	1	Digikey: 17-2314820-6-ND, Mouser: 855-M22-7140342[4]

Notes:

1. We recommend gold plated pin headers for reliability.

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2. Single row sockets are also readily available from numerous sources though good quality ones are not that expensive. We recommend you use good quality, gold plated headers and sockets. Note: for using Ethernet with a socketed Teensy 4.1, care must be taken to select 2 mm and 2.54 mm headers and sockets that are similar in height.
3. The 2 mm header and socket must be matched to the Teensy 4.1 socket and header height.

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Bill of Materials for PCBA

Designator	Footprint	Comment
C1, C10, C13, C2, C20, C28, C8, C9	402	100n
C11	402	1uF
C12	1206	1nF
C14	CP_Elec_6.3x7.7	47 uF
C21	402	68pF
C22, C26	805	22uF
C27	603	10uF
C3, C5, C6, C7	402	10n
C4	603	1uF
D1, D2, D3, D4	SOT-23	BAT54SSOT-23
D24	D_SMA	D_Schottky
D5, D6	LED_805	KT-0805G
IC1, IC10, IC2, IC3, IC4, IC5, IC6, IC7, IC8, IC9	EL3H7-G	EL3H7
IC11, IC12, IC17	TSSOP-20	74AHCT541A
IC13, IC14	DBV-6	74LVC2G17DBV
IC16	DBV-SOT-23	74LVC1G32DBV
IC18	SOIC-8	LM358D
IC19	SOIC-16	ULN2003AD
IC20	SOT-23-6	TPS54202DDCR
IC21	SOT-23-5	74LVC1G07
L2	SWPA6045S100MT	10uH
Q1, Q2	SOT-23	BSS138
R1, R10, R11, R2, R25, R3, R4, R5, R6, R7, R8, R9	1206	3K3
R12, R14, R15, R19	402	47K
R13, R17, R18, R21	402	1K
R16	805	1M
R20, R22	402	2.2K
R23, R24, R26, R27, R28, R29, R32, R33, R34, R35	402	10K
R30, R36, R39, R40	805	0R
R49	402	100K
R51	402	13K7
RV1	TrimPot TC33X	BOURNS_TC33-2

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